



Customer Shopping Behavior — Executive Summary

Analysis of 3,900 transactions to reveal spending patterns, segments, product performance, and subscription behavior to guide strategic decisions.

Dataset at a Glance

Size & Scope

3,900 rows × 18 columns covering demographics, purchases, behavior, and shipping.

Key Features

Age, Gender, Location, Item, Category, Amount, Season, Rating, Subscription, Shipping, Discounts.

Missing Data

37 missing values in Review Rating (imputed by category median).



Data Preparation & Feature Engineering

01

Cleaning

Loaded with pandas, inspected with `df.info()` and `.describe()`; standardized column names to `snake_case`.

02

Imputation

Imputed Review Rating nulls using product-category median.

03

Engineered Features

Created `age_group` and `purchase_frequency_days`; dropped redundant `promo_code_used`.

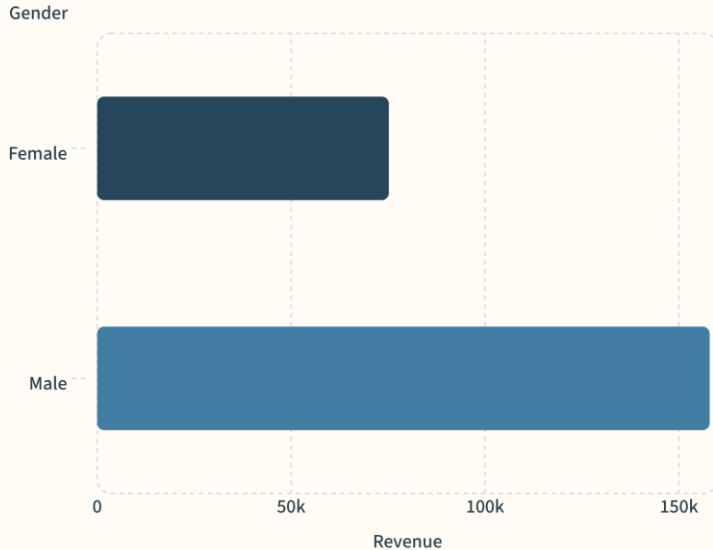
04

DB Integration

Loaded cleaned DataFrame into PostgreSQL for SQL analysis.



Revenue & Demographics (SQL Findings)



Male customers generated higher total revenue (157,890) vs Female (75,191). Use this to inform targeted campaigns and inventory allocation.



Top Products & Ratings



Gloves

Average rating: 3.86 — top-rated product in the dataset.



Sandals

Average rating: 3.84 — strong performer in footwear.



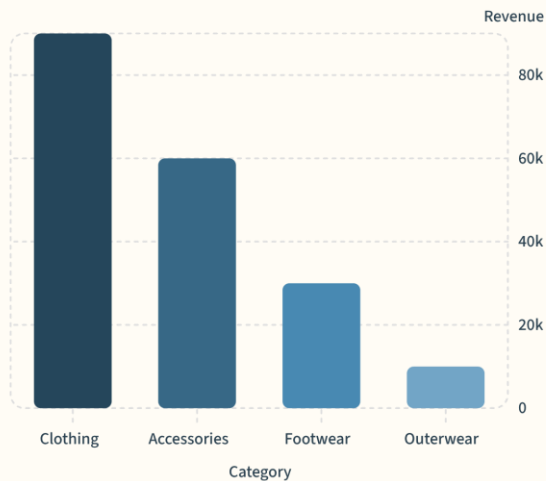
Boots

Average rating: 3.82 — consistent customer satisfaction.

Also notable: Hat (3.80) and Skirt (3.78).

Sales & Revenue by Category

Clothing leads revenue and sales volume. Prioritize merchandising, promotions, and inventory for Clothing while supporting high-margin Accessories.





Customer Segmentation & Loyalty

3.9K

Customers

\$59.76

Avg Purchase

3.75

Avg Rating

Segments: Loyal (3,116), Returning (701), New (83). Loyal customers form the majority—focus retention and rewards.

Repeat buyers (>5 purchases) are more likely to subscribe: repeat buyers split — No: 2,518; Yes: 958. Leverage loyalty programs to convert returning to loyal.

Discounts, Shipping & Subscription Insights

Discounts: Top discount-dependent items include Hat (50%), Sneakers (49.66%), Coat (49.07%), Sweater (48.17%), Pants (47.37%).

Shipping: Avg purchase — Standard: \$58.46; Express: \$60.48.

Subscriptions: Subscribers (1,053 customers) avg spend \$59.49, total revenue \$62,645; Non-subscribers (2,847) avg \$59.87, total \$170,436.



Dashboard & Visual Story



Interactive Power BI dashboard visualizes: subscription share (Yes 27% / No 73%), revenue and sales by category and age group, and key KPIs (3.9K customers, \$59.76 avg purchase, 3.75 avg rating).

Use dashboard to explore filters: subscription_status, gender, category, shipping type.

Business Recommendations

Boost Subscriptions

Promote exclusive benefits and targeted offers to increase subscriber conversion.

Loyalty Programs

Reward repeat buyers to move Returning customers into Loyal segment.

Review Discount Policy

Balance promotion uplift with margin control; reassess high-discount products.

Product Positioning

Highlight top-rated and best-selling items (e.g., Gloves, Sandals, Clothing) in campaigns.

Targeted Marketing

Focus on high-revenue age groups (Young Adult, Middle-aged) and express-shipping users for premium offers.